

Program: Diploma in Civil and Rural Engineering	
Course Code: 3369	Course Title: Building and sanitation Engineering Drawing Lab
Semester: 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Impart basic knowledge of the principles of building planning and drawing
- To introduce the fundamentals of Civil Engineering drawing and understand the principles of planning.
- Enable the students to prepare submission drawings

Course Prerequisites:

Topic	Course code	Coursename	Semester
Engineering Graphics		Engineering Graphics	S1

Course Outcomes:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Gain knowledge of basic principles of building drawing	12	Applying
CO2	Apply principles of building planning and prepare line plans by following the building rules	14	Applying
CO3	Sketch sanitary, plumbing, rain water harvesting and electrical layouts of a building	10	Applying
CO4	Prepare submission drawings	5	Applying
	Lab Tests	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4			3				

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline

On completion of the course student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Gain knowledge of basic principles of building drawing		
M1.01	Discuss the Conventions (as per IS 962) and symbols for different materials and Graphical symbols for Structural elements and sanitary and electrical installations	2	Understanding
M1.02	Sketch the Wall footing with step, RCCcolumn footing, combined footing	3	Applying
M1.03	Draw the front elevation, sectional plan and sectional elevation of a Panelled door and glazed window. Dog legged RCCstair	4	Applying
M1.04	Sketch the plan and elevation of Gable and Hip roof.	3	Applying

CO2	Apply principles of building planning and prepare line plans following building rules		
M2.01	Discuss the Principles of planning for Residential and Public building- Aspect, Prospect, Orientation, Grouping, Privacy, Elegance, Flexibility, Circulation, Furniture requirements, Sanitation, Economy. Discuss the Introduction to Rules and bye-laws of sanctioning authorities for construction work KMBR, KPBR, CRZ Act, NBC, KMMCR). Discuss Space requirement and norms for minimum dimension of different units in the residential and public buildings as per Kerala Building Rules Define coverage, covered area, floor space index, Parking area, Ventilation, Headroom, Mezzanine floor, Set back, Corner Plot, Small plots, Huts, Highrise building. Draw the site plan as per principles of planning.	4	Understanding
M2.02	Sketch line plans for residential building of minimum three rooms including water closet(WC), bath and staircase as per principles of planning.	3	Applying
M2.03	Sketch Line plans for Institution building such as School building and Hostel building as per principles of planning.	3	Applying
M2.04	Draw the plan, section and elevation of one BHK residential building as per principles of planning.	3	Applying
M2.05	Study the basics of electrical layout and fire and safety plan of a institution building	1	Understanding
	Lab Test-I	2	
CO3	Sketch service plans		
M3.01	Study of plumbing and sanitary fittings	2	Understanding
M3.02	Sketch Plumbing layout showing water supply and sanitary fixtures of a residential building (one BHK Residential Building)	3	Applying

M3.03	Draw the drainage plan ,water supply layout and layout of bath and lavatory of a building	3	Applying
M3.04	Sketch the plan and section of Rain water harvesting system	2	Applying

Text/Reference:

T/R	BookTitle/Auth or
T1	Dr. A Achuthan, Dr.Balagopal& T S Prabhu: Building Planning and Drawing ;SpadespublishersShah.M.G.Kale,CM,Patki,S.Y.,BuildingDrawing,Mcgraw HillPublishing
T2	Shah.M.G.Kale,CM,Patki,S.Y.,BuildingDrawing,McgrawHillPublishing
R1	MalikandMayo,CivilEngineeringDrawing,ComputechPublication Ltd
R2	Swamy,Kumara;Rao,N,Kameshwara,A.,BuildingPlanningandDrawing,CharotarPublication, Anand.
R3	PunmiaB.C.,andJainA.K.,BuildingConstruction ,FirewallMedia.
R4	Mohan,G.MuthuSobha:Principles ofArchitecture,OxfordUniversityPress
R5	IS962:Codeofpractice forArchitecturalandBuildingDrawings
R6	KeralaMunicipal BuildingRule
R7	KeralaPanchayath BuildingRule
R8	NationalBuildingCode
R9	KeralaMinorMineralConcessionRules
R10	Harmonized guidelines and space standards for barrier free built environment forpersons with disabilityand elderlypersons, 2016

OnlineResources:

Sl.No	WebsiteLink
1	https://nptel.ac.in/
2	http://egyankosh.ac.in/ https://www.coursera.org/learn/construction-cost-estimating
3	https://swayam.gov.in/